engineering 4 us all

Student Handout

Name: Ninja Cat **Date:** 12/10

Element I: Testing, Data Collection and Analysis

To ensure reliability and usability, the prototype was tested for durability, removability, compact design, push-only functionality, and force.

Durability

We replicated five years of usage by mounting a prototype wheel on a treadmill. The treadmill operated at a constant speed of one rotation per second for ~2.77 hours ($10000\ spins\ x\ 1\ Seconds\ =\ 10000\ Seconds$, $10000\ Seconds\ /\ 3600\ Seconds\ =\ 2.77\ hours$) which simulated continuous wear. Inspections throughout the test revealed no signs of cracking, deformation, or fatigue. We also dropped the device from 1m, with no further damage. These results not only validated the material's resilience but also provided confidence that the product would withstand extended use without compromising performance.

Removability

Testing involved 10 participants in a controlled environment, each tasked with removing the modification under timed conditions. All participants succeeded within one minute, with an average time of 10 seconds. User feedback collected during post-trial interviews highlighted the design's simplicity.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Compact Design

The compact design of the prototype was a key feature. By minimizing its size to 2x8x3 inches, we ensured that 98% of the cart's original loading space was preserved, surpassing the initial target of 90%. This also only took up space on the bottom level which saved a ton of space on the top level. Simulated loading tasks

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demonstrated that users could maneuver and load the cart without interference. The design achieved its intended goal of maximizing utility without sacrificing space.

Push-Only Functionality

We planned to integrate a remotely controlled Boombot to automate cart movement, eliminating the need for manual pushing. This feature addressed a critical need by prioritizing both convenience and user health.

Testing for force to pull the cart

We tested multiple springs one by one, adding more until we had four spring scales. Then, we used them to pull the cart and calculated the total force needed by summing the readings from each scale. The number of Newtons required to pull the 3D printer and the cart was approximately 15 Newtons. We investigated the printer's model and estimated how many Newtons it would take to pull 500 pounds.

3d Printer Model Snapmaker 0350T Weight 28.0 kg → 61.7294 lbs

The desired weight to pull is 500 lbs, $500\ lbs \div 61.7294\ lbs/printer \approx 8.099\ Printers$

Each 3d Printer requires ~15 Newtons to pull and 500 pounds is about 8.099 printers.

$8.099\ Printers \times \sim 15\ Newtons$

About ~121.498 Newtons to pull

The wood prototype also required 3 wheels to stay balanced to pull the cart without constantly tipping forward, the current prototype is 3 pounds and needed ~50 pounds to stay stable on the ground.

The prototype was thoroughly tested for durability, removability, compact design, and push-only functionality, with results confirming its reliability and usability. Durability tests showed no damage after simulated years of use and drops, while

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removability testing demonstrated ease of use, with participants removing the modification in under a minute. The compact design saved space, and force testing revealed that approximately 121.5 Newtons of force were required to pull a 500-pound load using the 3D printer, with additional weight needed for stability.